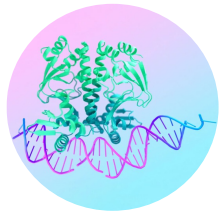
SciLifeLab

# AI agents in action







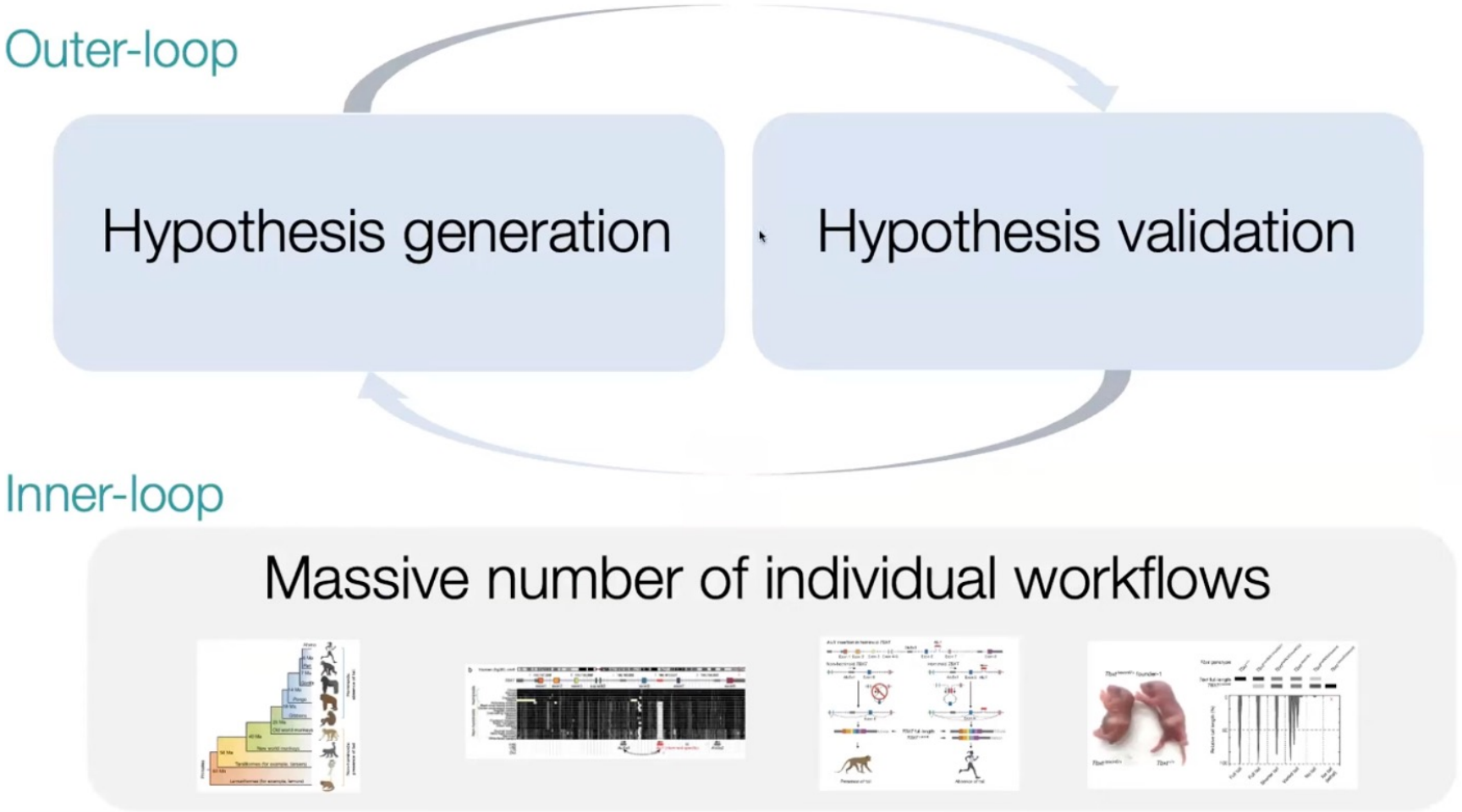
Mahbub Ul Alam
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SciLifeLab Data Centre
mahbub.ul.alam@scilifelab.se





# Life science research is a loop of reasoning and actions

**Every discovery cycles between thinking and doing.**




[ 1: Biomni talk: https://www.youtube.com/watch?v=kOIrpy8V7GM f ]

# What we want from a life science AI agent



**Six things that turn a chatbot into a research collaborator.**

| Capability | Details |
|---|---|
| Adapt | Plug in tools from genomics, drug discovery, clinical data, … any domain |
| Unify | Orchestrate workflows across multiple domains |
| Evolve | Add new skills as research advances |
| Efficient | Automate repetitive data tasks |
| Reproducible | Transparent, executable record of every step |
| Scalable | Tackle complex, multi-step research at scale |

# Two ways to put an agent to work

**Pick by how well an existing agent fits your problem.**




**Path 1, Adopt:** your task matches a ready-made agent? Use it. Fastest route, nothing to build.

**Path 2, Build:** need your own data, tools, or workflow? Assemble your own.

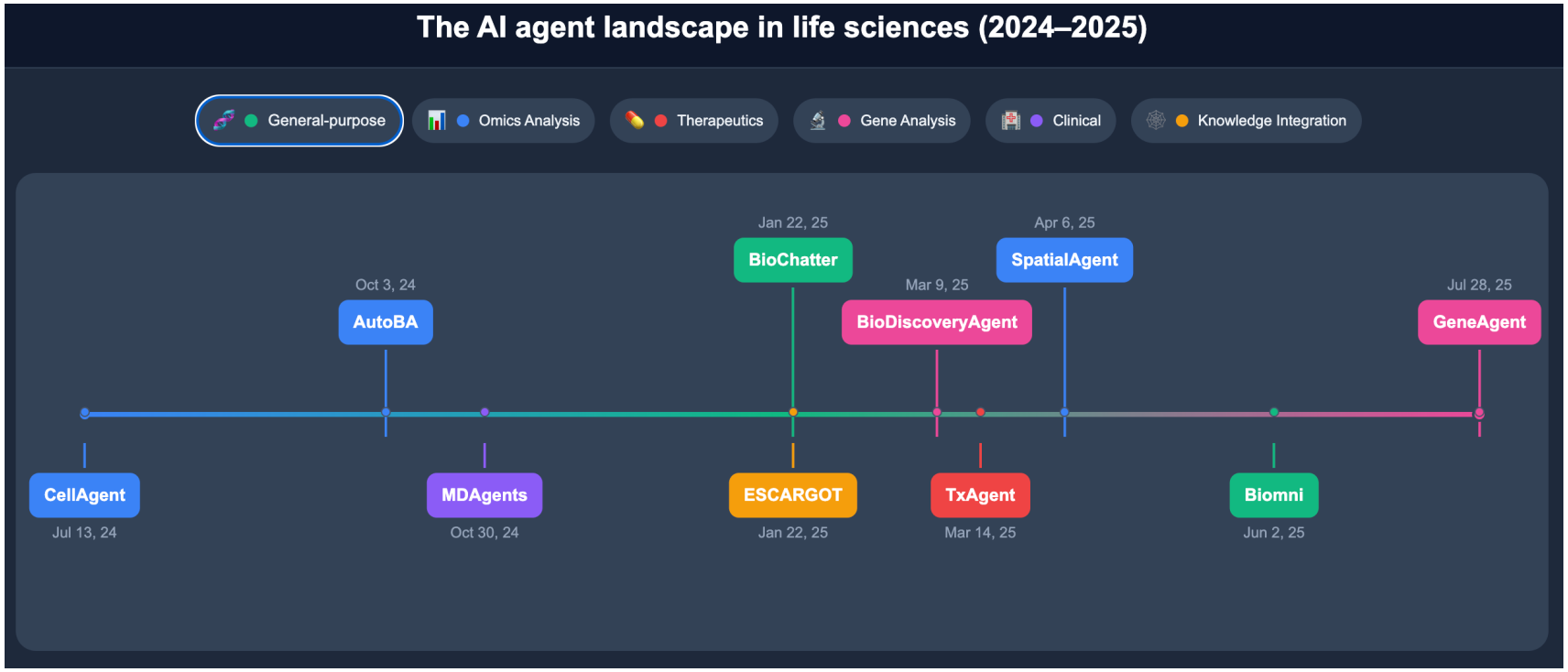
**The dividing line is the tools:** ready-made agents **bring their own**, a **custom agent wraps yours** (your scripts, databases, APIs, even a whole deployed app).

# Path 1, find the right agent for the job

**Chances are, someone already built one for your task.**

**Tool 1**

**shorturl.at/dlpN6**

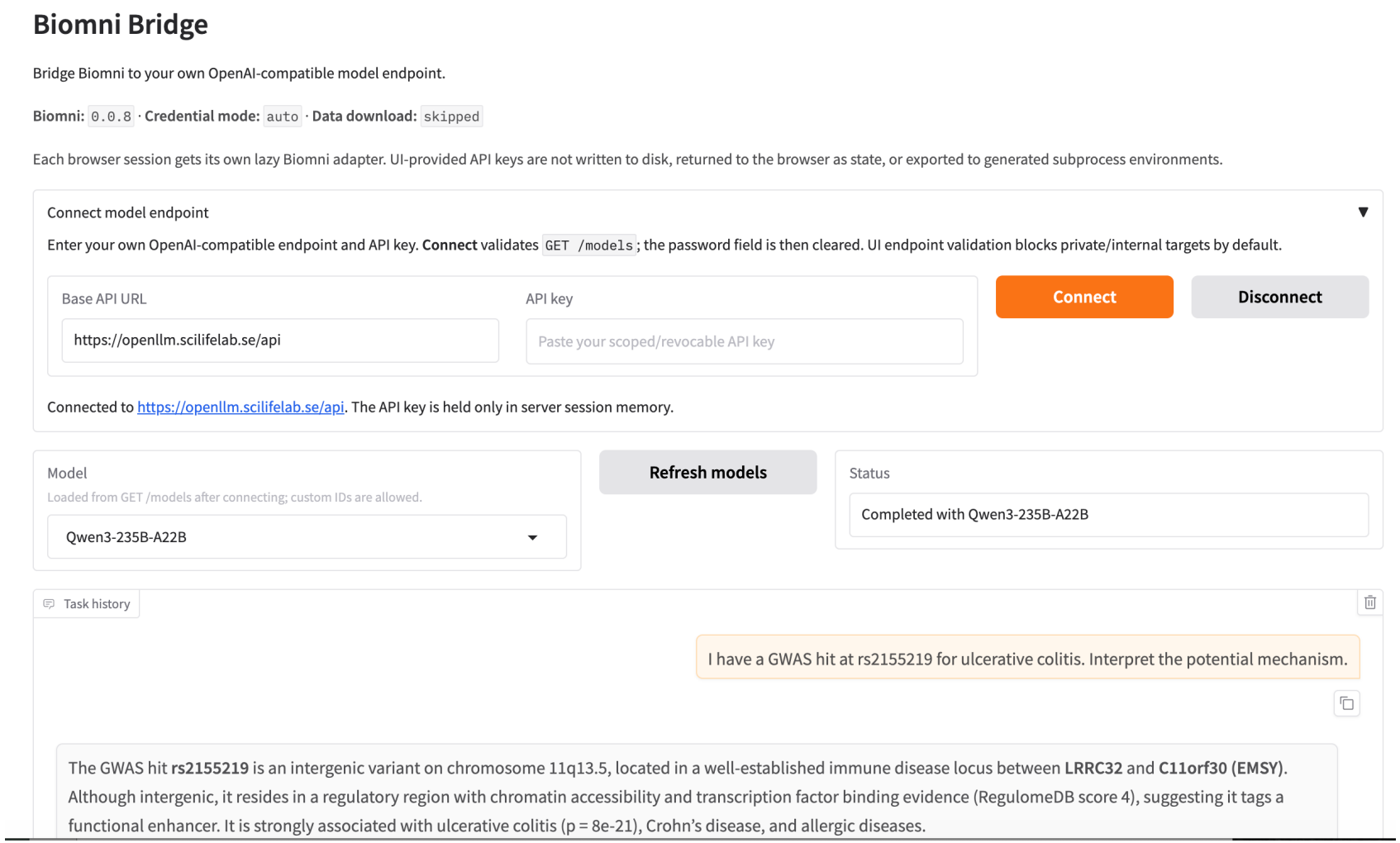



**Tool 2**

**shorturl.at/lcmru**

**Explore two AI agent landscape tools**

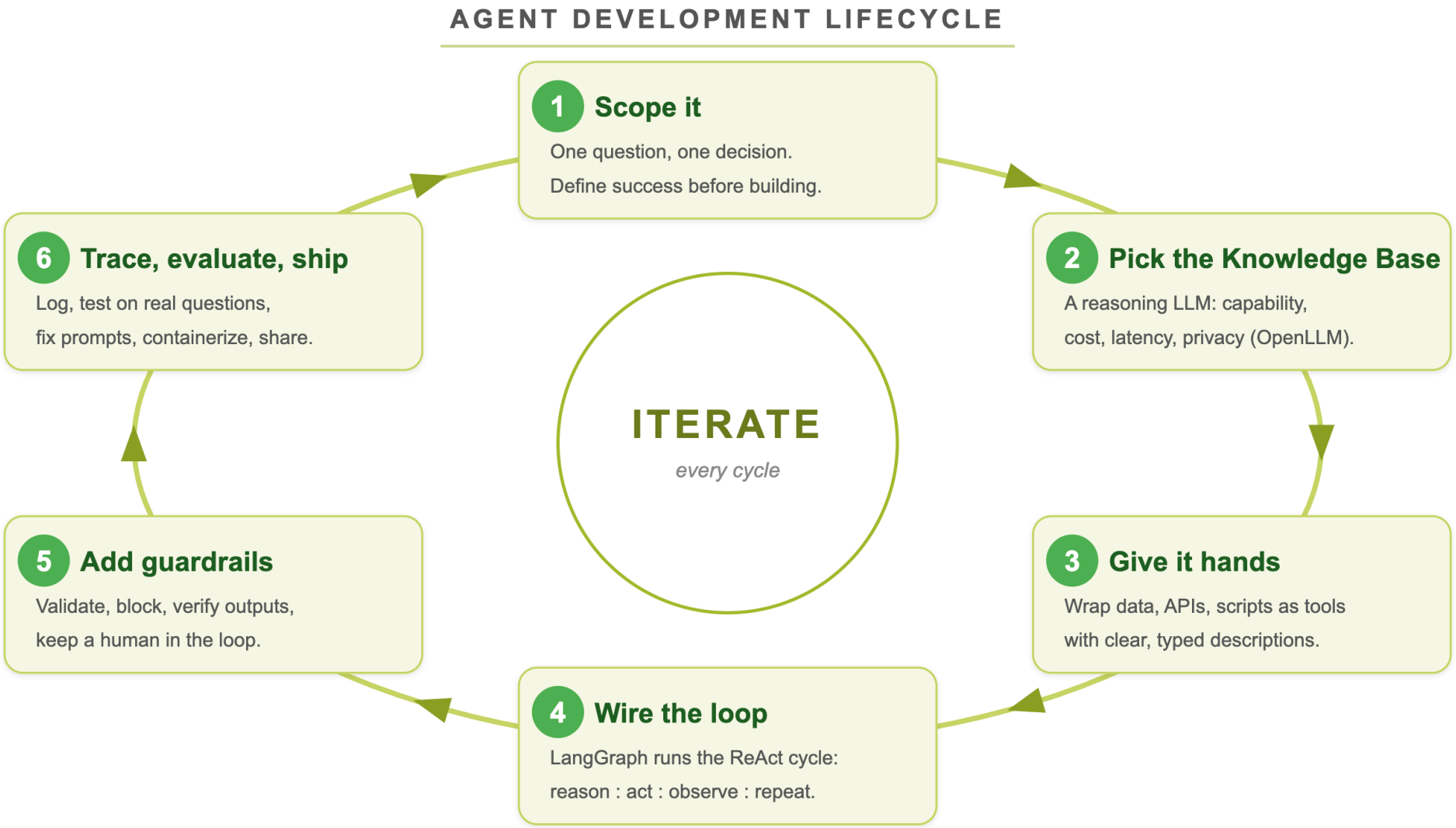
# Path 1, Adopt: Biomni Bridge




**Biomni Bridge**

**shorturl.at/LQINt**

# Path 2, Build: a typical development workflow




**Same loop as research: define, equip, reason, act, check, repeat.**

# Path 2, The builder's toolbox

**You have choices. Today we pick one stack and build end to end.**

- **Frameworks:** LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen. **Today: LangGraph (Session 1).**
- **The knowledge base:** any reasoning LLM, commercial or open. **Today: the SciLifeLab OpenLLM pilot. (Session 1 & 2)**
- **Sharing** + deployment: MCP to expose tools, Docker to package, SciLifeLab Serve to host. **Today: MCP (Session 2).**

# Agent-ready research infrastructure: **serve.scilifelab.se/apps**

## FROM ONE TOOL TO A PLATFORM

**SHAMSUL**

**Chest X-ray pathology**

+ Grad-CAM · LIME · SHAP · LRP

*the one tool you just saw*

wrap as a tool

**156+**

public research apps on SciLifeLab Serve

serve.scilifelab.se/apps

## WHAT'S ALREADY DEPLOYED

| | | | |
|---|---|---|---|
| Drug Discovery | Proteomics | Genomics & Transcriptomics | Spatial Biology & Imaging |
| Metabolomics & Exposomics | Clinical & Public Health (demo) | Ecology & Environment | Plant Biology |
| Metagenomics | Knowledge Graphs | Education | Data & General Tools |

**Connect and call the tool, not download and hope it runs.**

*Each one a candidate to become agent-ready.*

# Thank you! 😊



Next up: **Hands-on Session 1**

build your own agent with LangGraph and ReAct.

**The future belongs to those who adapt!** 😊